Fiscal Year:	FY 2025	**Task Last Updated:**	FY 12/08/2025
PI Name:	Vitaterna, Martha Ph.D.		
Project Title:	Effects of Partial Gravity on Multi-system Mammalian Physiology: Microbiome and Related Systems		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Microbiology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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PI Web Page:			
City:	Evanston	**State:**	IL
Zip Code:	60201-2919	**Congressional District:**	9
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2017 Space Biology HERO 80JSC017N0001-BPBA Topics in Biological, Physiological, and Behavioral Adaptations to Spaceflight. Appendix C
Start Date:	09/01/2020	**End Date:**	08/31/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 08/31/2026 per F. Hernandez/ARC (Ed., 10/2/25). NOTE: End date changed to 08/31/2025 per NSSC information (Ed., 9/2/24). NOTE: End date changed to 08/31/2024 per NSSC information (Ed., 1/16/24). NOTE: Title was changed from original proposal when awarded--proposal title was "Impact of the Martian Solar Day and Gravity on Microbiota in Mice: Mechanisms and Multi-System Physiology" (Ed., 9/1/21)		
Key Personnel Changes/Previous PI:	Stefan Green, PhD, CoInvestigator, has moved from University of Illinois at Chicago (UIC) to Rush University. His role has not changed. Dr. George Chlipala, CoInvestigator, is no longer with the project as our work with the analysis core at UIC has ended. Dr. Xuanyi Lin (Northwestern) and Dr. Adit Chaudhary (Rush) have joined the team working on bioinformatic analyses.		

COI Name (Institution):	Jiang, Peng Ph.D. (Northwestern University, Evanston) Green, Stefan Ph.D. (Rush University) Turek, Fred Ph.D. (Northwestern University, Evanston)
Grant/Contract No.:	80NSSC20K1519
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The gastrointestinal microbiota plays a key role in mammalian immune, metabolic, and psychological health. The spaceflight environment harbors challenges to gut microbial communities and increases markers of gastrointestinal inflammatory diseases. Our prior evaluation of gut microbiota using samples from a previous Rodent Research mission (RR-1) and comparison to data from mice on a shuttle mission (STS-135) have shown reproducible spaceflight-specific shifts in gut microbiota, although the specific influence of microgravity, or partial gravity, relative to other spaceflight factors or stressors remains unknown. Particularly, it has not been demonstrated whether partial gravity is sufficient to restore the spaceflight-induced shift in microbial community structure. Hence, understanding of the impact of partial gravity on the microbiota will help us prepare for future human space missions that may include exposure to micro- and partial gravity. We hypothesize that partial gravity may mitigate some, but not all effects of spaceflight on the gut microbiota. To test this hypothesis, we propose to utilize the Japan Aerospace Exploration Agency (JAXA) Mouse Habitat Unit (MHU) onboard the International Space Station (ISS)'s Kibo module, which provides the ability to house mice under artificial gravity from 0-1 G. We will evaluate the gut microbiota, as well as metabolic, circadian, sleep/wake, and gut/immune function parameters in mice exposed to microgravity, partial (0.33G or 0.67G) gravity, or 1G artificial gravity. Results from the proposed studies will allow for a better understanding of the relationships between the gut microbial community and mammalian physiology, addressing a key knowledge gap regarding the risk of adverse health effects due to host-microorganism interactions under conditions relevant to travel to other planets.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The knowledge gained regarding the impacts of gravity and other spaceflight stressors on the gut microbiome, as well as how changes in the gut microbiome are matched (or not) by changes in other physiological systems, will provide insight into the role played by the gut microbiome in mammalian adaptation to stress. This will enable microbiome-focused strategies for mitigating the health impacts of challenges both in spaceflight and Earth.
Task Progress:	See Cumulative Bibliography for publications resulting from this NASA funding (Ed., 12/8/25).
Bibliography Type:	Description: (Last Updated: 12/08/2025)
Articles in Peer-reviewed Journals	Kremsky I, Pergerson R, Justinen S, Stanbouly S, Willey J, Fuller CA, Takahashi S, Vitaterna MH, Bouxsein M, Mao X. "Artificial gravity attenuates the transcriptomic response to spaceflight in the optic nerve and retina." Int J Mol Sci. 2024 Nov 9;25(22):12041. https://doi.org/10.3390/ijms252212041 ; PubMed PMID: 39596110; PubMed Central PMCID: PMC11593819 , Nov-2024